GINO – Technical Overview and Validation

An open-source, mid-fidelity framework for hydroelastic flutter of composite lifting surfaces · UVLM + Capytaine BEM

LV · 27 October 2026

Note: synthesis of the author's MSc thesis (March 2026) [2] and of a manuscript prepared for Ocean Engineering, not yet peer reviewed [1]. Results are for early-stage design. Full references at the end.

Nature of this document

What it is

- A technical presentation of the author’s own research, not a peer-reviewed publication.
- Main source: the manuscript prepared for Ocean Engineering, draft of 25 August 2026, not yet peer reviewed [1].
- Background: the MSc thesis, Politecnico di Torino, March 2026 [2].
- Numbers are those reported in [1]. Figures are taken from [1]; where marked “adapted”, curves outside the scope of this document were removed.
- Results are intended for early-stage design.

Original contribution (GINO)

Coupling architecture, injection of sectional matrices and BEM operators, unified apparent-mass bookkeeping, solution routes, validation campaign. Code: [3].

Third-party open-source tools used

SONATA [4] and ANBA4 [5] for sections; SHARPy linearised UVLM [6], [7]; Capytaine BEM [8].

External reference data

Model 2T experiment by Besch & Liu [9]; Goland wing [10] with the SHARPy reference solution [11].

Why GINO

The problem

- Composite lifting surfaces are lighter and more strongly bend–twist coupled. In water the added mass is of the same order as the structure, so flutter becomes a design constraint.
- High-fidelity FSI is accurate but suited to targeted checks, not parametric studies.
- Existing reduced-order tools: OpenFAST [15] and GEBTAero [16] rely on 2D strip aerodynamics; AeroFlex [17] has no cross-section preprocessor; DCFoil [18] does not assemble a fully populated stiffness; SHARPy [6] exposes no entry point for added mass and radiation damping.

What GINO adds

Sectional matrices as first-class objects

Fully populated 6×6 matrices from SONATA/ANBA4 enter the beam without translation: a new layup gives a new flutter point in one automated loop.

BEM hydrodynamics at modal level

Capytaine A(ω) and B(ω) enter the coupled equations in place of the kernel's own apparent mass, removed by a single rule.

Open source and reproducible

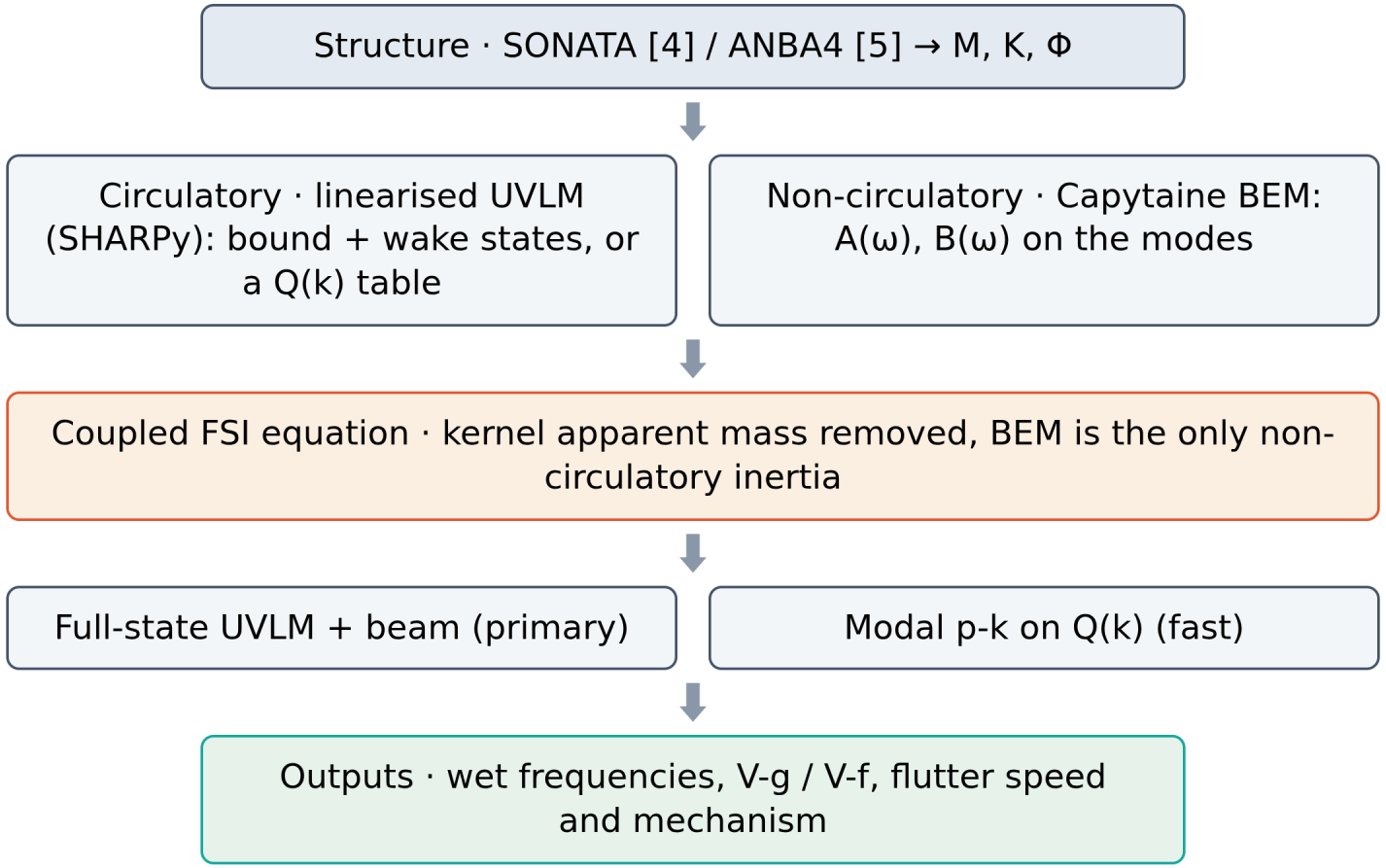
Code, benchmark cases and figure scripts at github.com/lorebasket/GINO [3].

From thesis to paper

The thesis [2] built the toolchain; the paper [1] made it physically consistent in water and tested it against a measured flutter point.

item	thesis (March 2026)	GINO today (paper)
Structure	SONATA/ANBA4 → 1D beam FEM	Timoshenko beam with fully populated 6×6 matrices
Water inertia	Capytaine A(∞) replaces the kernel apparent mass	one rule for every route: kernel apparent mass removed, BEM A(ω) and B(ω) inserted
Solution	p-k with mode tracking; Roger RFA	full-state UVLM + beam (primary); modal p-k (fast)
Water check	NACA0003 in still water [19]: dry +0.2 %, wet +11.4 %	Besch & Liu Model 2T [9]: measured flutter, +6 %

Architecture



How to read it

- The structure is the only upstream dependency: UVLM and BEM interpolate the mode shapes Φ onto their own meshes.
- New layup → new M, K, Φ → the UVLM Q(k) and the BEM A, B are rebuilt; the two fluid operators stay independent of each other.
- Sectional matrices enter at beam assembly, hydrodynamics enter at modal level: the two injection points.

Structure: from layup to beam

- The foil is cut into spanwise slices; SONATA [4] turns each into a 2D finite-element mesh with shape, plies, fibre angles and webs.
- ANBA4 [5] (open source) solves the Saint-Venant problem: 12 polynomial solutions (6 rigid, 6 deformable) from small linear systems instead of 3D elasticity.
- Output per section: fully populated 6×6 stiffness S, mass matrix, shear centre and centre of gravity.
- Timoshenko beam: $K_e = \int B^T S B \, dx_1$, so every coupling term in S reaches the FE equations.
- Modal reduction keeps the first n_m mass-normalised modes Φ.

Why it matters

Layup-induced bend–twist coupling and the elastic-axis / CG offset are exactly the terms that set the flutter mechanism.

Bonus

The same Saint-Venant basis rebuilds 3D stress fields for composite failure criteria.

Limit

Sections are locally straight and prismatic: strongly curved or swept foils need finer spanwise slicing.

Fluid: two operators, two physics

Circulatory · linearised UVLM (SHARPy [6])

- Vortex lattice with explicit free wake, linearised about the steady state (Maraniello & Palacios [7]).
- States are bound and wake circulations: wake memory is exact, no rational fit of the lift deficiency.
- Incompressible, which suits water.
- Used as the full state, or reduced to a modal Q(k) table for p-k.
- Q(k) depends on the mode shapes: a new layup costs one k-sweep.

Non-circulatory · Capytaine BEM [8]

- Mode shapes are interpolated onto the wetted panel mesh: each mode becomes a radiation motion of the body.
- Capytaine solves the radiation problem and returns A(ω) and B(ω) directly in modal coordinates.
- True 3D geometry: thickness, volume, free surface, spanwise and cross-modal coupling.
- Strip theory instead projects a 2D sectional added mass after the fact.

In water the mass ratio is of order one [20]: added mass is not a correction, it reorganises the spectrum and can decide which branch flutters.

The key step: one water mass only

- A potential-flow kernel carries its own flat-plate apparent mass: the $-k^2B_2$ behaviour of Re Q(k).
- In air that is the physical inertia. In water it is the wrong one: no thickness, no 3D volume, no free surface.
- Adding the BEM on top counts the water twice; keeping the kernel term alone models it badly.
- GINO's rule: identify and remove the kernel apparent mass, so the BEM operators are the only non-circulatory terms, whatever the solution route.

p-k route

$B_2 = -\mathrm{Re}\ Q(k_{b2}) / k_{b2}^2$

$\mathrm{Re}\ Q_m(k) = \mathrm{Re}\ Q_m^{raw}(k) + k^2\ B_2$

Coupled equation

$M_{eff} = M_m + A_h(\omega)$

$C_{eff} = C_m + B_h(\omega) - \tfrac{1}{2}\rho(Vb/k)\ \mathrm{Im}\ Q_m$

$K_{eff} = K_m - \tfrac{1}{2}\rho V^2\ \mathrm{Re}\ Q_m$

Full-state route

UVLM apparent-mass feedthrough ($\dot{\gamma}$ block) removed; Capytaine A(ω), B(ω) injected on the modal mass and damping blocks.

Delta form

UVLM table intact + (A_BEM(ω) − A_UVLM,nc) in mass + B_BEM(ω) in damping. Matrices rebuilt independently equal A_BEM to 1e-17.

Two solution routes

Full-state eigenanalysis · primary

- State = structural modes plus every bound and wake circulation.
- Eigenvalues of the coupled system at each speed: no k sampling, no p-k iteration.
- Physical roots separated from the wake continuum by beam-state participation, chained across speed by MAC.
- Resolves heavily damped, wake-dominated coalescence partners that a low-order modal basis misses.
- Cost: eigenproblems orders of magnitude larger.

Modal p-k [12] · fast alternative

- Q(k) sampled once on a reduced-frequency grid and cached to disk.
- Inner iteration on $k = \omega b/U$ until consistent; outer sweep over speed.
- Hybrid initial guesses, deferred correction, MAC tracking with predict-and-select [13].
- Exact only near neutral stability: approximate for heavily damped branches.
- A full V-g / V-f sweep is a handful of $n_m \times n_m$ eigenproblems.

Flutter is read where $g = \sigma/|\omega|$ crosses zero (positive = unstable). A Roger RFA route [14] is also available; in water its fit tends to over-stiffen bending and advance coalescence.

Validation criteria

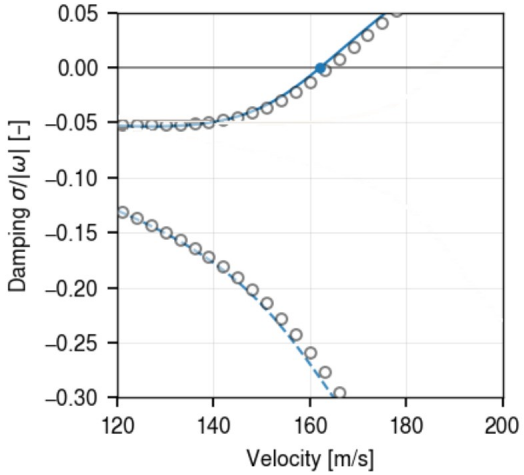
criterion	case	reference	outcome
Coupled chain, in air	Goland wing [10]	SHARPy full-order [11]	163.7 vs 163.8 m/s (0.1 %), 11.3 Hz
Flutter in water	Model 2T, Besch & Liu [9]	NSRDC experiment	19.2 vs 18.1 kn (+6 %), 6.7 vs 6.4 Hz
Mechanism	Model 2T, BEM vs strip	experiment	torsion 6.7 Hz (BEM); spurious 4.4 Hz coalescence (strip)
Convergence	SONATA sections, UVLM grid	finest grid	6–10 sections; 30×12 panels, Q(k) error 1.0 %
Robustness	10 inputs one at a time, Model 2T	experiment	18.1 kn inside every input’s envelope

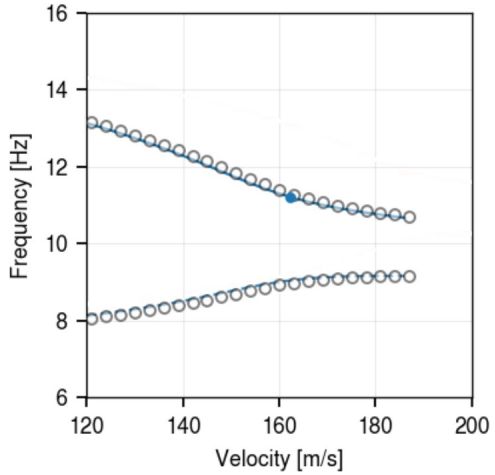
Verification in air: Goland wing

Set-up

- Uniform cantilever, classical bend–torsion benchmark.
- Goland wing [10], same configuration as the SHARPy example [11]: 16×32 panels, wake 10 chords, $\rho = 1.02$ kg/m³.
- Full-state UVLM, 4 modes.
- Same apparent-mass bookkeeping as in water, for consistency.

Purpose: check the coupled solver chain, not the marine physics.





source	V_F [m/s]	f_F [Hz]
SHARPy full-order	163.8	11.3
GINO, UVLM full-state	163.7	11.3

0.1 % on flutter speed; same mechanism (torsion branch near 11 Hz, coalescing with the rising bending branch)

Validation in water: Model 2T

19.2 kn

GINO flutter speed, UVLM + BEM (experiment 18.1 kn, +6 %)

6.7 Hz

flutter frequency on the torsion branch (experiment 6.4 Hz)

0

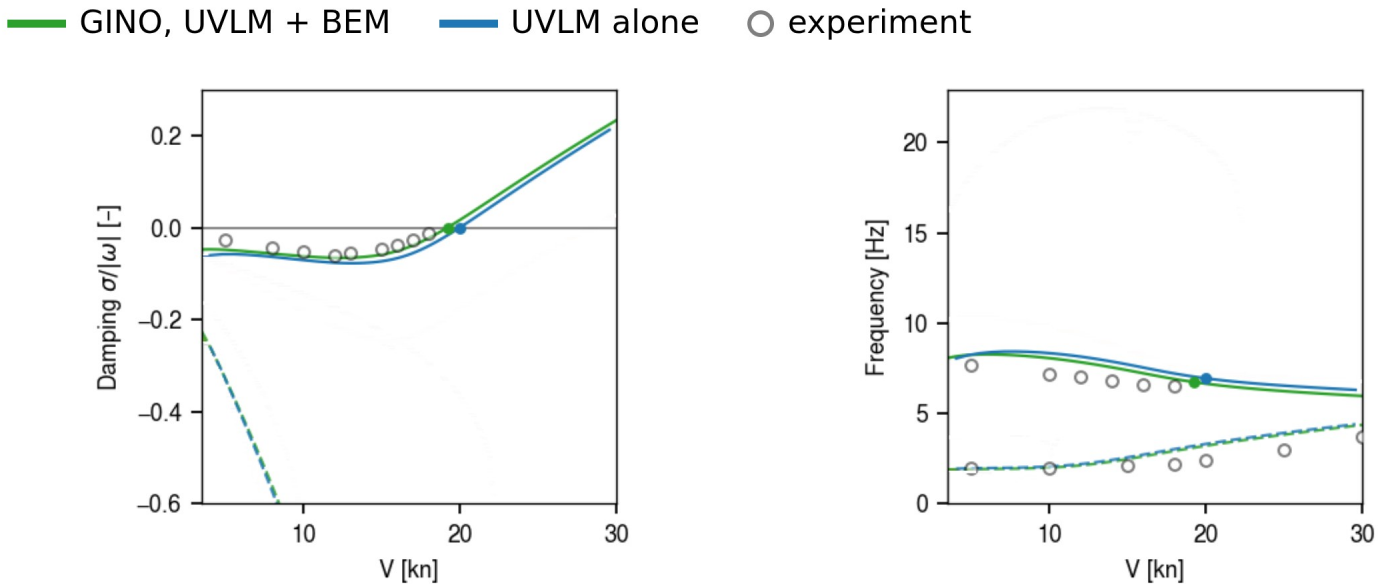
structural calibrations: nominal stiffness throughout

- Besch & Liu (NSRDC, 1973) [9]: surface-piercing, blunt-based strut with a tip pod (configuration G); measured flutter 18.1 kn (NSRDC) / 18.4 kn at 6.4 Hz, torsion-dominated.
- Demanding marine case: mass ratio of order one, geometry piercing the free surface, concentrated pod added mass.
- Model: beam on the shear-centre axis with offset CG axis, lumped pod mass, BEM meshes of strut and pod; 4 wet modes; Rayleigh $\zeta = 5$ % on the first two.
- Same chain without BEM (stock SHARPy UVLM at water density): 20.0 kn at 6.9 Hz (+10 %). Modal p-k with BEM: 20.2 kn at 6.9 Hz.

Model 2T: V-g and V-f

Reading

- Both UVLM solutions start at 8.1–8.2 Hz against a measured wet torsion of ≈7.9 Hz, and follow the measured descent toward flutter.
- BEM lowers the flutter point by 0.8 kn and 0.2 Hz versus the UVLM alone.
- On this case the circulatory kernel sets the answer; BEM refines the speed and, next slides, fixes the mechanism.

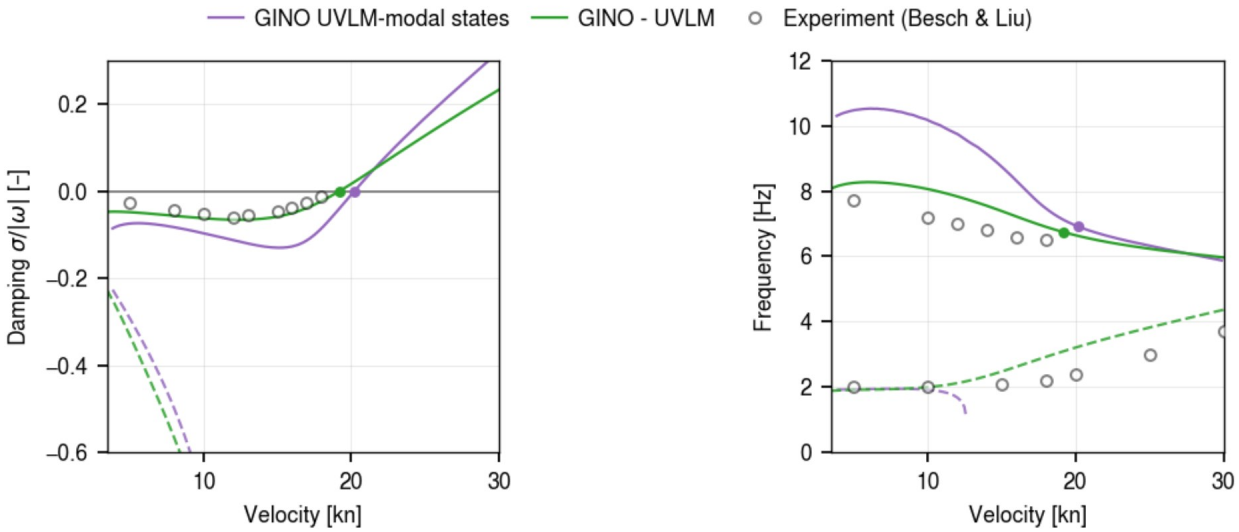


Solid: torsion branch · dashed: bending branch · dots: predicted flutter points · adapted from [1]

Full-state versus modal p-k

route	V_F [kn]	f_F [Hz]
full-state	19.2	6.7
modal p-k, 4 modes	20.2	6.9
experiment	18.1	6.4

- Same hydro-aerodynamic model; the modal route pays about 1 kn for its speed.
- Causes: basis truncation, and p-k evaluates the operator on the imaginary axis, exact only near neutral stability.
- Both routes predict flutter above the measured speed.

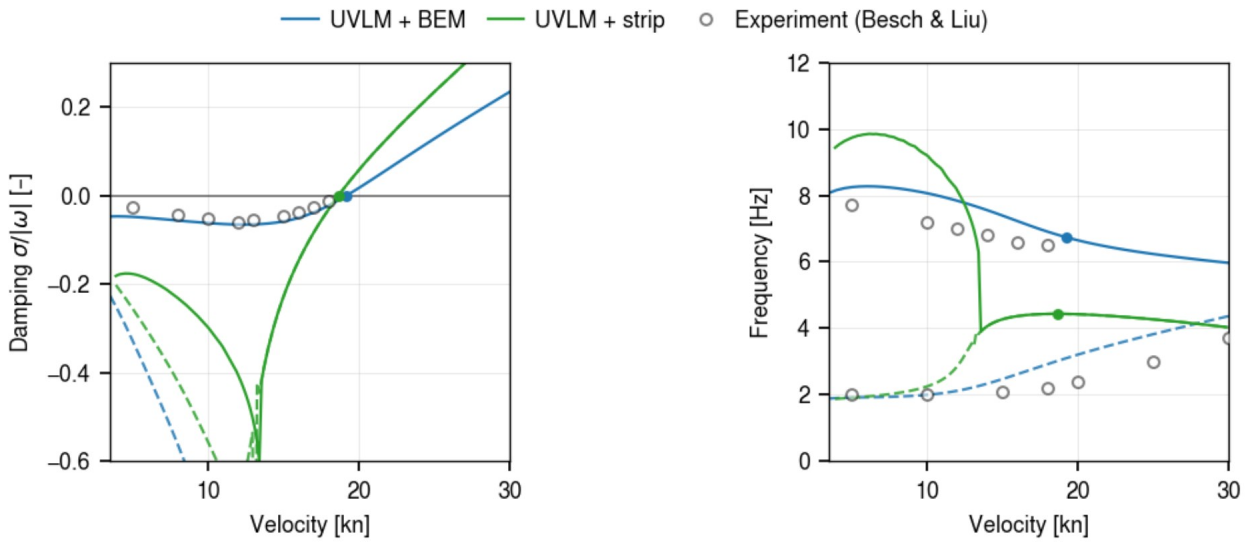


Purple: modal p-k · green: full-state · circles: experiment · from [1]

BEM versus strip theory: same speed, different physics

added mass	V_F [kn]	f_F [Hz]
Capytaine BEM	19.2	6.7
strip theory	18.7	4.4
experiment	18.1	6.4

- UVLM kernel fixed; only the non-circulatory inertia changes.
- BEM: torsion-dominated crossing at 6.7 Hz, as in the test.
- Strip: early bend–torsion coalescence at 4.4 Hz, a mechanism the test does not show.
- The strip speed is closer to 18.1 kn for the wrong reason: a speed-only check would reward it.



Blue: BEM, full-state · green: strip, modal p-k · circles: experiment · from [1]

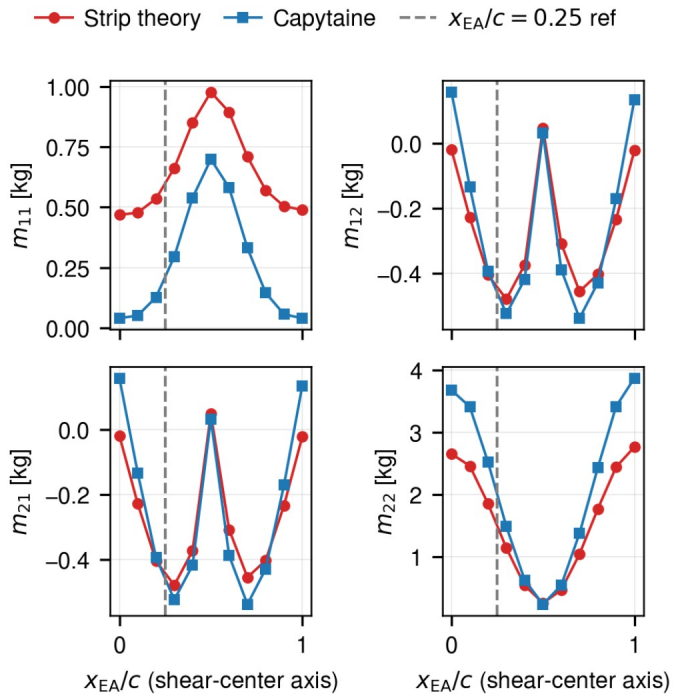
Why: the added-mass mismatch

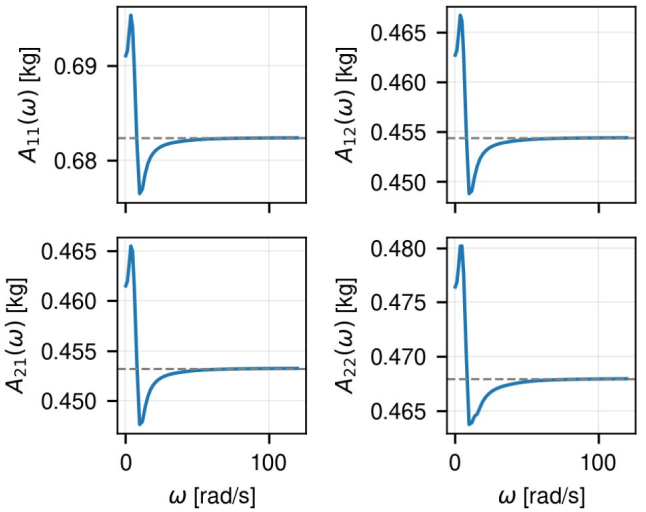
Elastic axis swept along the chord

- Strip and BEM agree near mid-chord, i.e. near the mass axis (x_CM/c = 0.512): gap ≈0.28 kg (Frobenius norm).
- Off the mass axis they diverge up to ≈1.2 kg: strip over-predicts bending m_{11} (~3× at x_EA/c = 0.25: 0.59 vs 0.20 kg) and under-predicts torsion m_{22} (1.49 vs 1.99 kg; 2.76 vs 3.87 kg at the trailing edge).
- Torsional and cross terms are the ones that set coalescence.

A(ω) on Model 2T

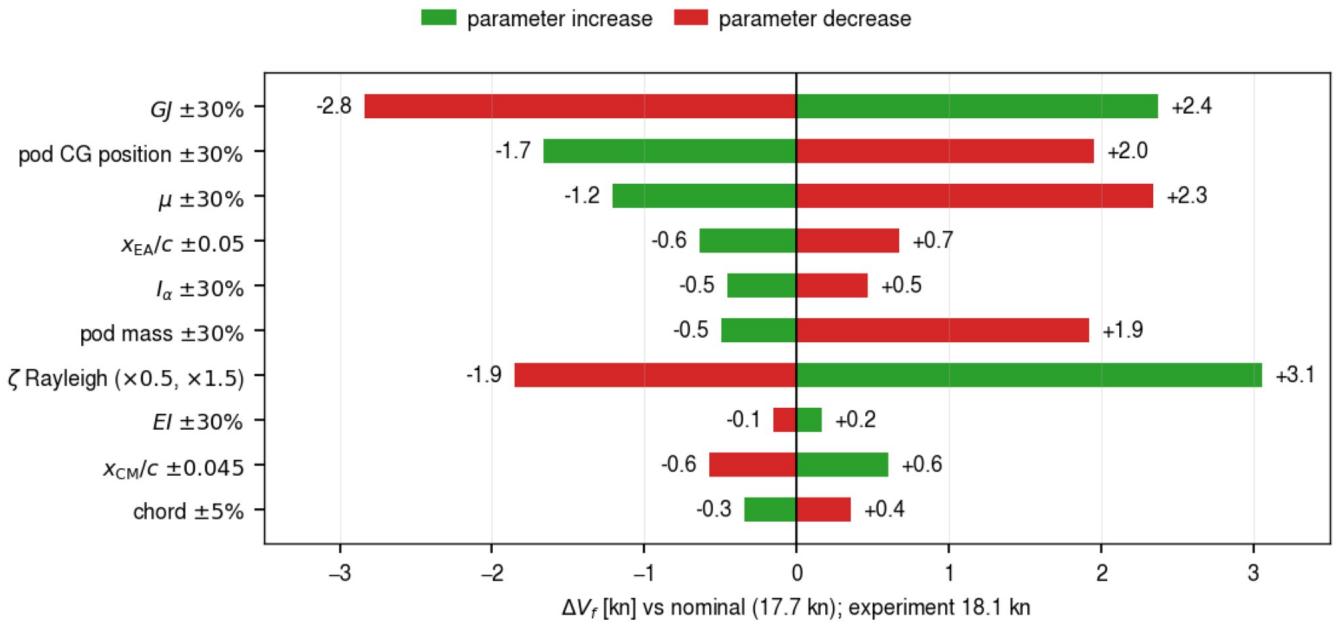
Free-surface transient below ~20 rad/s, constant from ~40–60 rad/s: GINO uses A(∞) in the flutter band.





Model 2T input sensitivity

- Baseline: modal p-k, UVLM, Capytaine; nominal Vf 17.7 kn.
- Ten inputs perturbed one at a time at ±10/20/30 %. The six weakest were pushed further (up to ±70 %, EI +100 %, damping ×0.1 to ×3): response stays linear, so the small effect is real.
- At ±30 %: Rayleigh damping +3.1/−1.9 kn, mass per span μ +2.3/−1.2, GJ −2.8/+2.4, pod CG −1.7/+2.0; axis positions ≈±0.6; Iα, chord, EI ≤0.5.
- Every input, damping included, can move Vf by knots; 18.1 kn lies inside the envelope of each one.



ΔVf at each input's most extreme valid level · green: increase · red: decrease · from [1]

UVLM grid convergence and cost

grid	panels	Q(k) error	case build	GAF table
10 × 4	40	9.2 %	2.8 s	0.1 s
20 × 8	160	3.1 %	2.7 s	1.1 s
30 × 12	360	1.0 %	10.8 s	10.8 s

Model 2T, error on the torsion-mode Q(k) at 6.4 Hz and 18.1 kn against the 40 × 16 grid. GAF table: 81-point k-sweep. Bold: production grid.

- Coarser grids carry 3–9 % error: not enough for quantitative flutter work on this geometry.
- BEM added-mass tables are computed once and cached.

30–72 s

one full V-g / V-f velocity sweep

On a laptop CPU

AMD Ryzen 5 5500U, 16 GB RAM, Ubuntu, single-process Python, no GPU.

A validated chain, link by link

1 · Structure SONATA + ANBA4

By the tool authors

Code-to-code vs VABS and NABSA on off-axis composite box beams; full 6×6 couplings [4].

In GINO

NACA0003 dry frequency within 0.2 % [2]; converged at 6–10 sections [1].

validated

2 · Fluid UVLM (SHARPy)

By the tool authors

Linearised UVLM state-space formulation [7], open source in SHARPy [6].

In GINO

GAF converged to 1.0 % (30 × 12); alone gives 20.0 vs 18.1 kn measured [1].

validated

3 · FSI chain in air Goland wing

Against

SHARPy full-order solution [10], [11].

Result

163.7 vs 163.8 m/s, 11.3 Hz: 0.1 %; same mechanism [1].

verified

4 · FSI chain in water Model 2T

Against

Besch & Liu experiment [9].

Result

19.2 vs 18.1 kn (+6 %), 6.7 vs 6.4 Hz, no calibration; right torsion mechanism [1].

validated

Verified = matches a reference code; validated = matches measurements or the tool's published validation. Water validation rests on one experimental campaign.

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[15] National Renewable Energy Laboratory, OpenFAST: open-source wind turbine simulation tool, github.com/OpenFAST/openfast (2024).

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